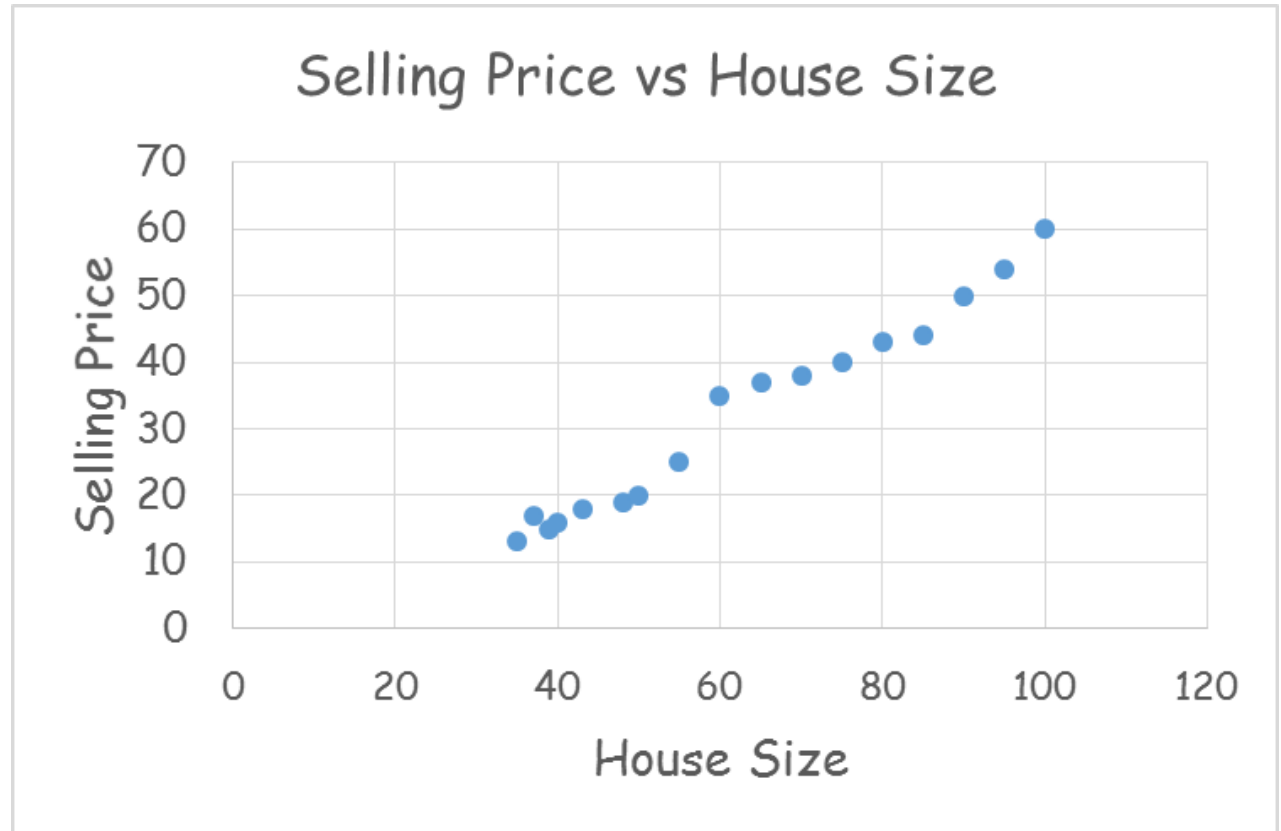


Simple Linear Regression

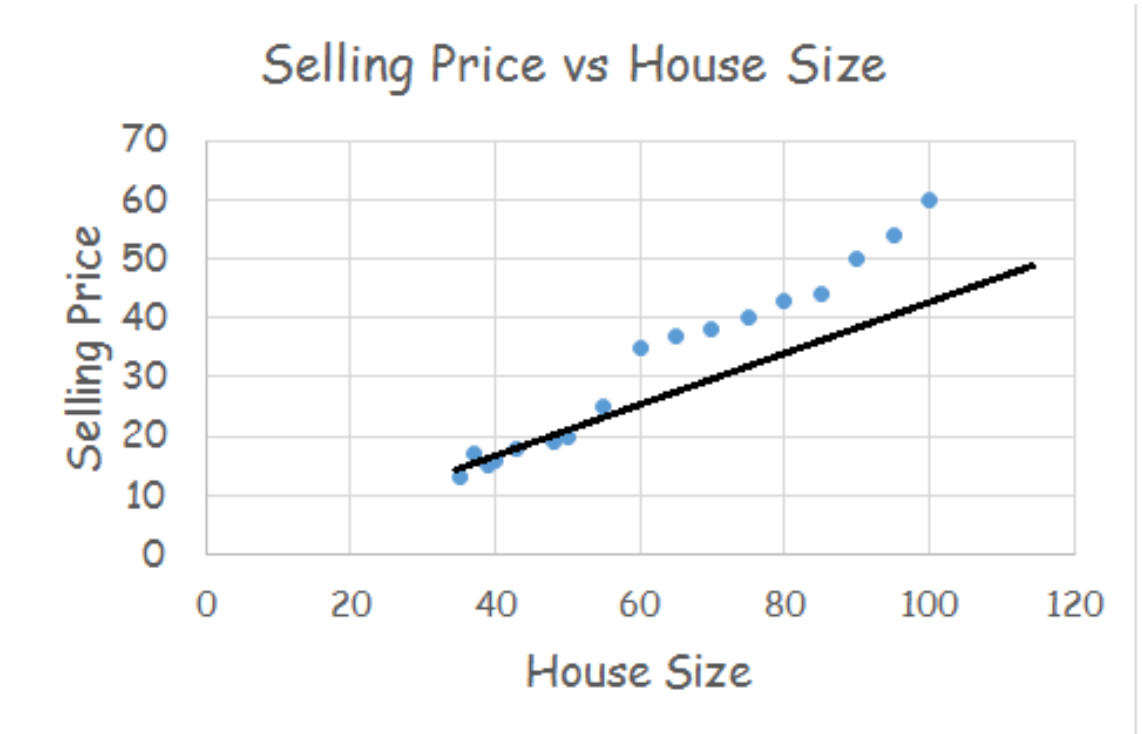
House Price Prediction

Area (SQM)	Selling Price (Lacs)
35	13
37	17
39	15
40	16
43	18
48	19
50	20
55	25
60	35
65	37
70	38
75	40
80	43
85	44
90	50
95	54
100	60



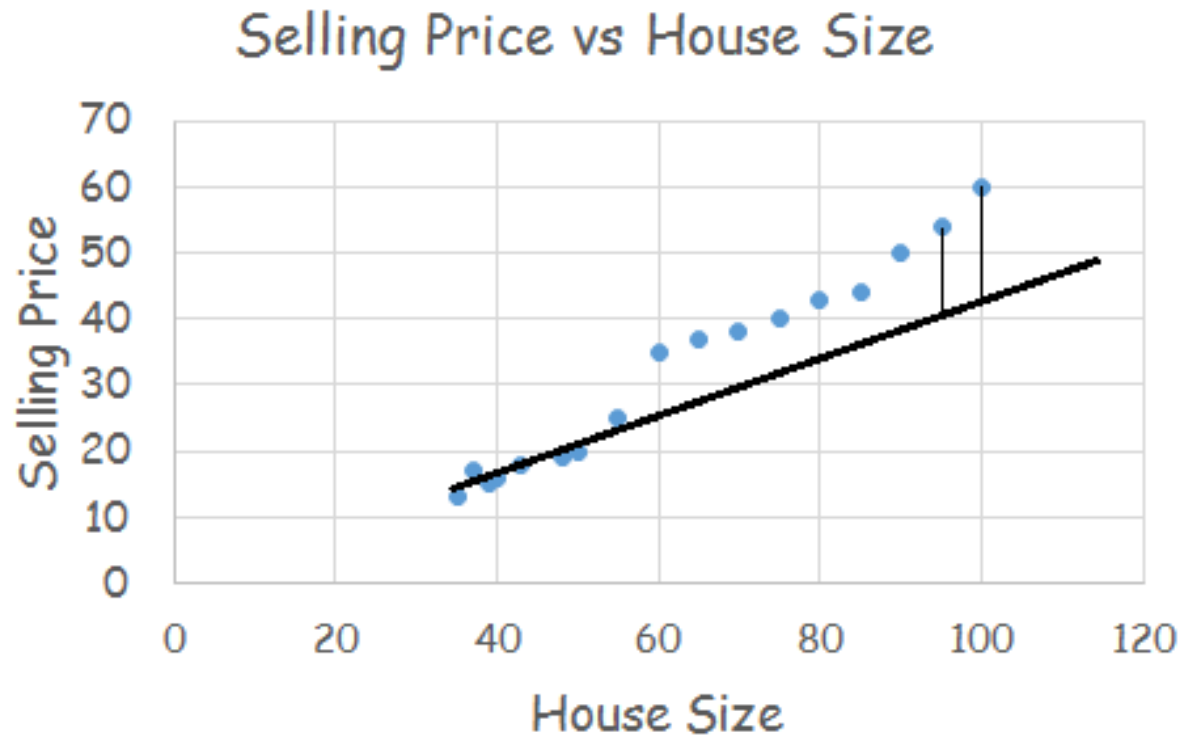
House Price Prediction

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55	25
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80	43
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95	54
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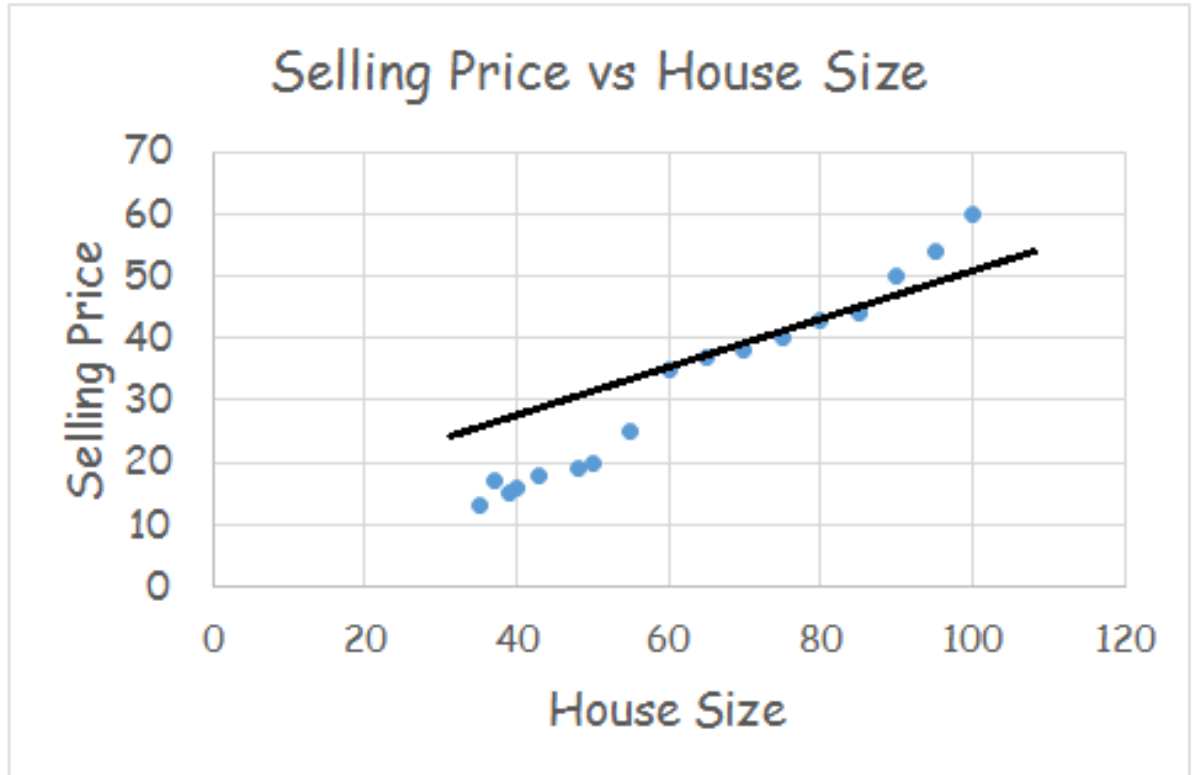
House Price Prediction

Area (SQM)	Selling Price (Lacs)
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55	25
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95	54
100	60



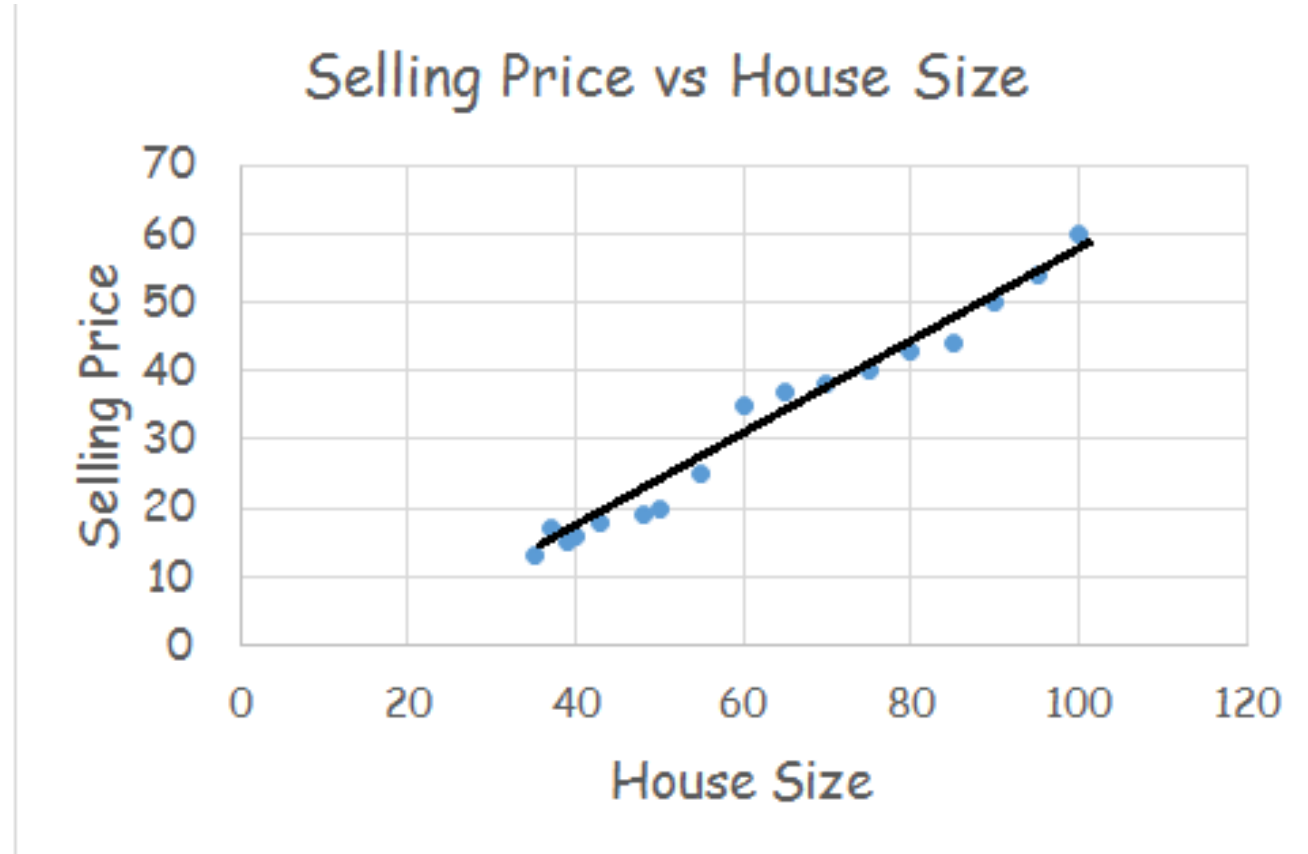
House Price Prediction

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House Price Prediction

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95	54
100	60



House Price Prediction

➤ Mathematically:

➤ Minimize $\sum_{i=1}^m (y_i - \hat{y}_i)^2$ $\hat{y} = h_{\theta}(x) = \theta_0 + \theta_1 x$

OR

➤ Minimize $(\frac{1}{2m}) \sum_{i=1}^m (y_i - \hat{y}_i)^2$

OR

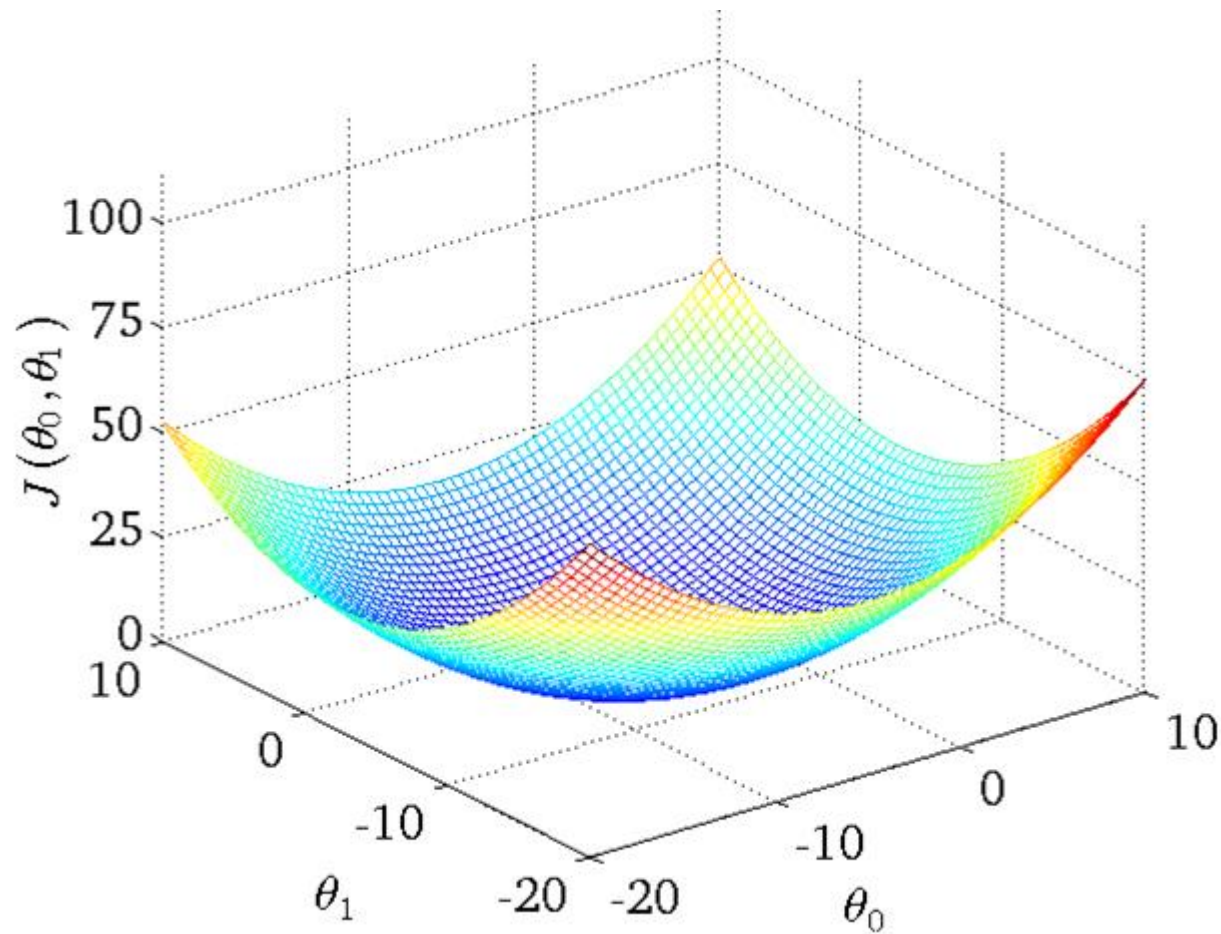
➤ $J(\theta_0, \theta_1) = (\frac{1}{2m}) \sum_{i=1}^m (y_i - \hat{y}_i)^2$

OR

➤ Minimize $J(\theta_0, \theta_1)$
 θ_0, θ_1

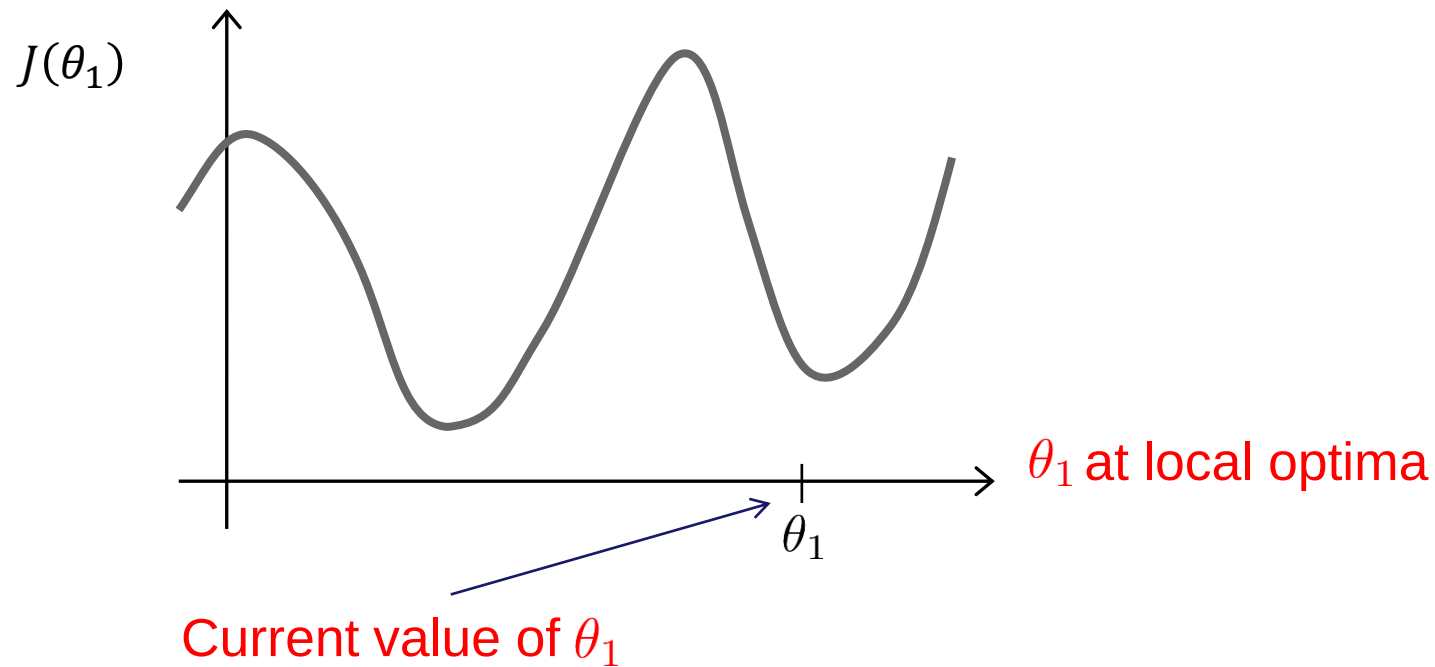
House Price Prediction

- Mathematically:



House Price Prediction

➤ Mathematically:



House Price Prediction

➤ Gradient Descent Algorithm

repeat until convergence {
 $\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$ (for $j = 0$ and $j = 1$)
}

Correct: Simultaneous update

$\text{temp0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$
 $\text{temp1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$
 $\theta_0 := \text{temp0}$
 $\theta_1 := \text{temp1}$

Incorrect:

$\text{temp0} := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$
 $\theta_0 := \text{temp0}$
 $\text{temp1} := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$
 $\theta_1 := \text{temp1}$

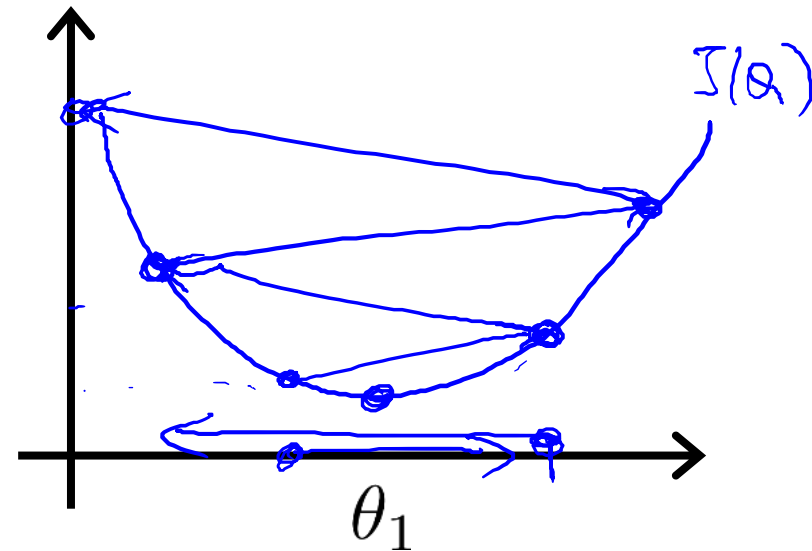
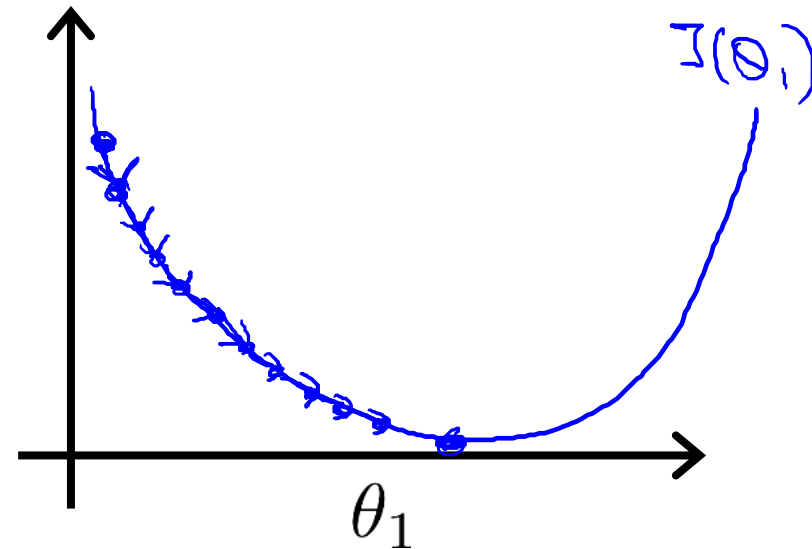
➤ $J(\theta_0, \theta_1) = \left(\frac{1}{2m}\right) \sum_{i=1}^m (y_i - \hat{y}_i)^2$

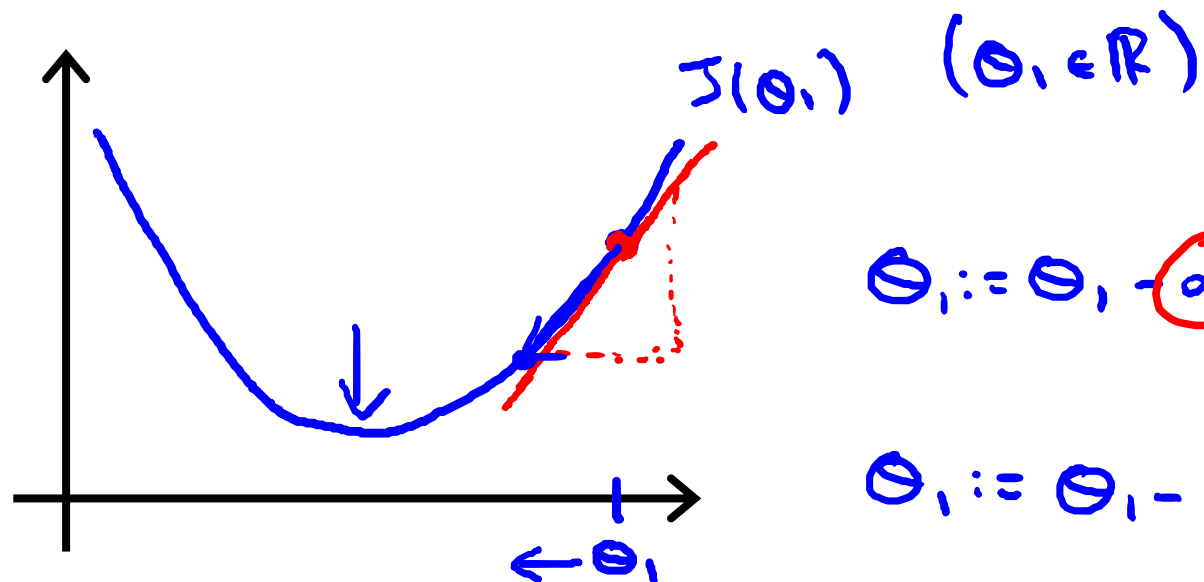
$$\hat{y} = h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_1)$$

If α is too small, gradient descent can be slow.

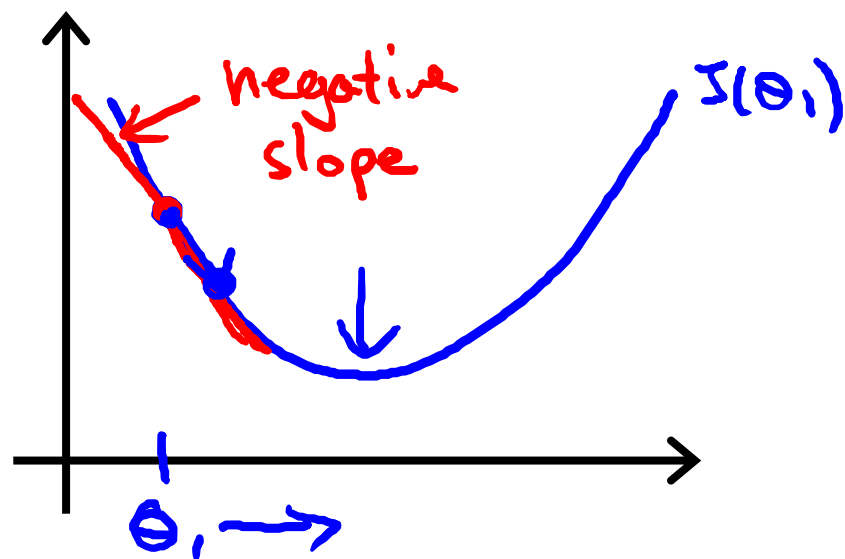
If α is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.





$$\theta_1 := \theta_1 - \underbrace{\alpha}_{\geq 0} \cdot \underbrace{\frac{\partial}{\partial \theta_1} J(\theta_1)}_{\geq 0}$$

$$\theta_1 := \theta_1 - \underline{\alpha} \cdot (\text{positive number})$$



$$\frac{\partial}{\partial \theta_1} J(\theta_1)$$

≤ 0

$$\theta_1 := \theta_1 - \underset{\uparrow}{\alpha} \underset{\uparrow}{(\text{negative number})}$$

House Price Prediction

➤ Gradient Descent Algorithm

Gradient descent algorithm

repeat until convergence {
 $\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$
 (for $j = 1$ and $j = 0$)
}

Linear Regression Model

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

House Price Prediction

➤ Gradient Descent Algorithm

$$\begin{aligned}\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{2}{2\theta_j} \frac{1}{2m} \sum_{i=1}^m (\underbrace{h_0(x^{(i)})}_{\text{red}} - y^{(i)})^2 \\ &= \frac{2}{2\theta_j} \frac{1}{2m} \sum_{i=1}^m (\underbrace{\theta_0 + \theta_1 x^{(i)}}_{\text{red}} - y^{(i)})^2\end{aligned}$$

$$j = 0 : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})$$

$$j = 1 : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

House Price Prediction

➤ Gradient Descent Algorithm

repeat until convergence {

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

}

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

update
 ~~θ_0 and θ_1~~
simultaneously

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\hat{y} = h_{\theta}(x) = \theta_0 + \theta_1 x$$

If $\theta_0 = 1$ and $\theta_1 = 0.5$?

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“Batch” Gradient Descent

“Batch”: Each step of gradient descent uses all the training examples.

$$\rightarrow \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

References

- Andrew Ng's slides on Linear Regression from his Machine Learning Course on Coursera.

Disclaimer

- Content of this presentation is not original and it has been prepared from various sources for teaching purpose.